4.1 Graphing Polynomial Functions

Name _____ Period ____ Date _____

Algebra II 4.1 · Student sheet

I can…

- identify polynomial functions and give their degree, type, and leading coefficient.
- evaluate polynomial functions.
- describe the end behavior of polynomial functions and graph them.

Vocabulary · fill in the missing word

Polynomial function — $f(x) = a_n x^n + \dots + a_1 x + a_0$ with _____ exponents

Degree — the _____ exponent

Leading coefficient — the coefficient of the term with the _____ exponent (in standard form)

End behavior — _____

Types — _____ (0), linear (1), quadratic (2), cubic (3), quartic (4)

Part 1 · End behavior sort

Function	Degree (even/odd)	Leading coefficient (+/−)	Left end	Right end
x^2				
x^3				
$-x^2$				
$-x^3$				
x^4				
$-x^5$				

Rule: _____

Part 2 · Notes and examples (fill in each note, then work the example)

Degree and leading coefficient: Degree = the greatest _____. Leading coefficient = coefficient of the term with the _____.

Example 1

Decide whether each function is a polynomial function. If it is, write it in standard form and give its degree, type, and leading coefficient.

a. $f(x) = 5x - 2x^3 - 8$ b. $g(x) = x^2 + 3x^{-1}$ c. $h(x) = 6 - \sqrt{2}\,x^4$ d. $k(x) = 5^x$

Example 2

Evaluate $f(x) = x^4 - 3x^2 + 2x$ when $x = -2$.

End behavior: Even degree: ends go the _____ way. Odd degree: ends go _____ ways.

Negative leading coefficient: A negative leading coefficient _____ the ends.

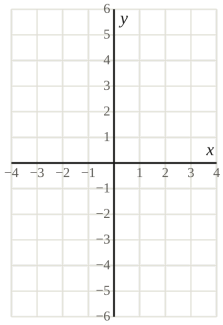
Example 3

Describe the end behavior.

a. $f(x) = -0.5x^4 + 3x^2 - 1$ b. $g(x) = 2x^5 - x$

Example 4

The graph shows a polynomial function. Is its degree even or odd? Is its leading coefficient positive or negative?



Part 3 · Partner practice

Partner 1

Write $f(x) = 4 - 7x + x^3$ in standard form. Give its degree, type, and leading coefficient.

Partner 2

Describe the end behavior of $f(x) = -3x^3 + x$.